

Unit 12, Lesson 10: Solving One-Step Multiplication and Division Equations in One Variable Using Manipulatives

Benchmark

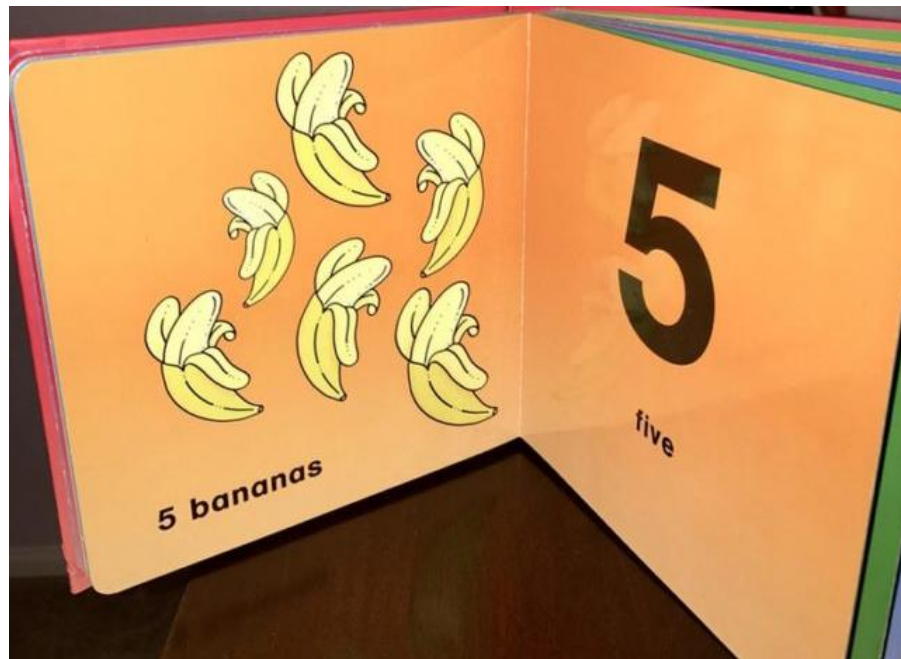
MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Target

- ☐ I can solve one-step multiplication and division equations using manipulatives.

12.10.1 Warm-Up: Math Fail

This children's book has published a "math fail."



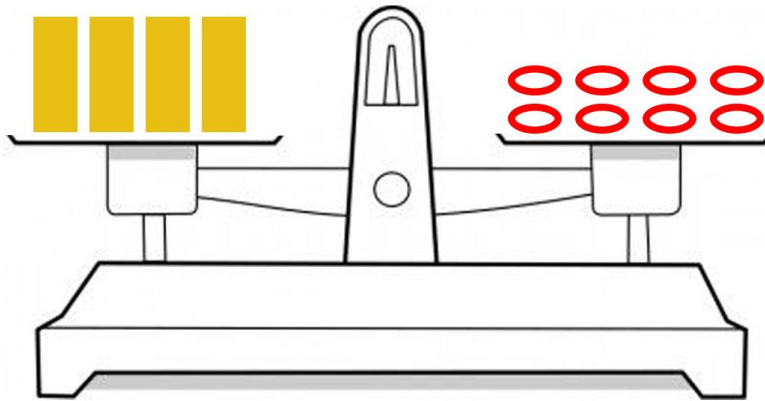
1. Explain the error shown in this image.

12.10.2 Exploration: Exploring Relationships

With your partner, complete the following.

1. Nadia made 8 friendship bracelets for her 4 best friends. Nadia wants to give all 8 of the bracelets away equally to each friend.
 - a. This situation can be modeled by the equation $4 \times \square = 8$. What operation is occurring in the equation?
 - b. Using mental math, how many bracelets will each friend receive?

This situation can also be represented using a balance scale. The rectangles represent Nadia's friends and the ovals represent the bracelets.



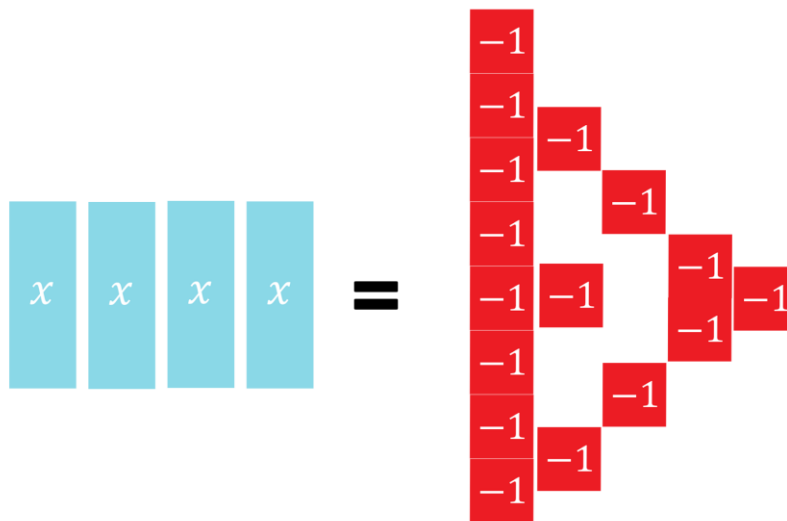
- c. Using the rectangles and ovals, sketch an image that demonstrates how many bracelets each friend will receive.
- d. What operation does your sketch demonstrate?

- e. How many bracelets will each friend receive?
- f. Discuss what you notice about the operation included in the equation and the operation you used to find the number of bracelets each friend will receive. Summarize your discussion.

Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

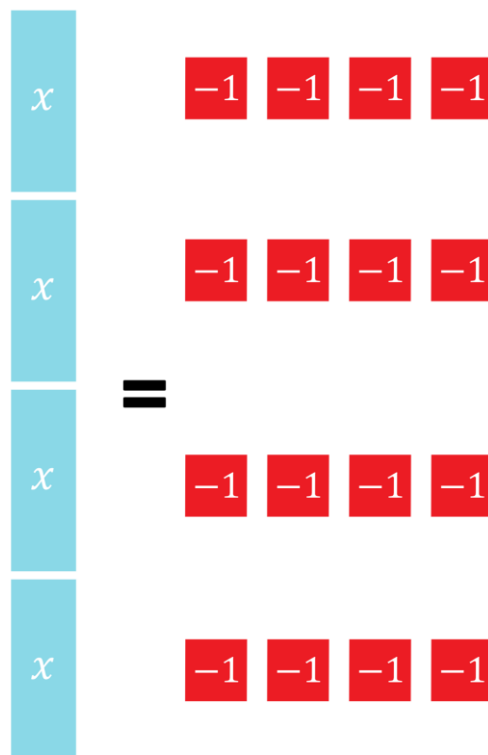
2. A model of the equation $4x = -16$ is shown using algebra tiles.

The  tile represents the variable, x , and the  represents the constant term where each tile is 1 unit.

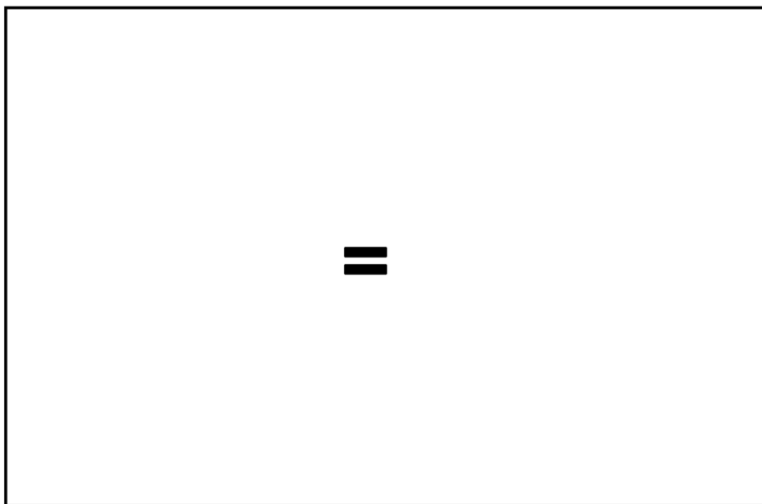


- a. What operation is being applied to the 4 and the x in the equation $4x = -16$?

- b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?



- c. Use algebra tiles to sketch the solution of the equation.



- d. What is the value of x ?

3. Inverse operations can be helpful in solving equations.
- a. Draw an arrow to match each operation on the left with its inverse operation on the right.
- | | |
|----------------|----------------|
| division | division |
| addition | addition |
| subtraction | subtraction |
| multiplication | multiplication |
- b. Explain what it means for operations to be inverses.
- c. Which inverse operations were used to solve the equations in this exploration?

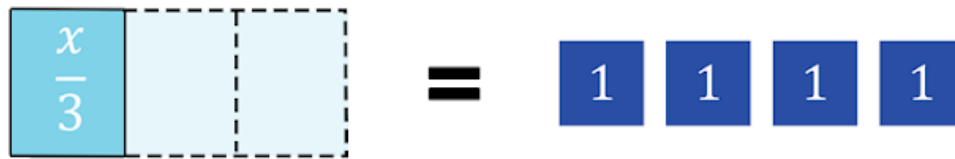
12.10.3 Guided Instruction: Modeling the Inverse of Division

Algebra tiles can also be used to model division equations.

1. The equation $\frac{x}{3} = 4$ can be rewritten as $x \div 3 = 4$.

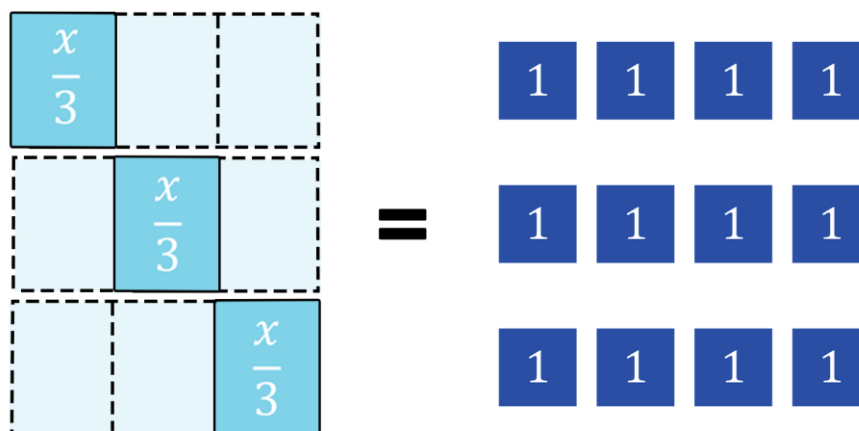
To represent division, show that x tile is being divided into equal parts. Sketch lines on the  tile to represent the fraction $x \div 3$.

This means when x is divided into 3 parts, each part is equal to 4. The model shown illustrates that $\frac{x}{3} = 4$.



- a. If each $\frac{1}{3}$ of x is equal to 4, discuss with your partner what the value of x is. As part of your discussion, include which operation can be used to determine the value of x . Summarize your discussion.

- b. The image models the relationship discussed that can be used to determine the value of x .



Explain how this representation can be used to determine the value of x .

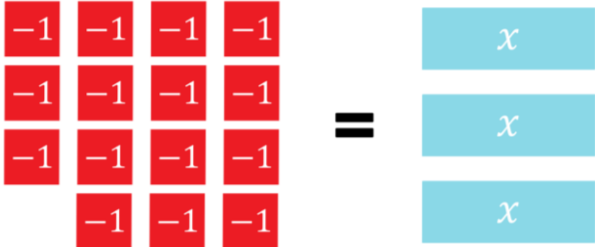
- c. What is the value of x ?

- d. Complete the statements by writing operations on each blank.

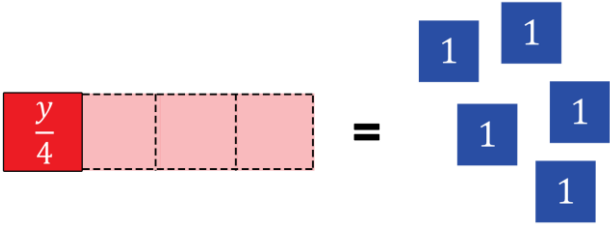
The original equation involved the operation _____. Its inverse operation, _____, was used to solve the equation.

12.10.4 Guided Practice: Solving Equations Using Manipulatives

1. Complete the table using algebra tiles to sketch the solution of $-15 = 3x$ and explain each step. The first row has been completed.

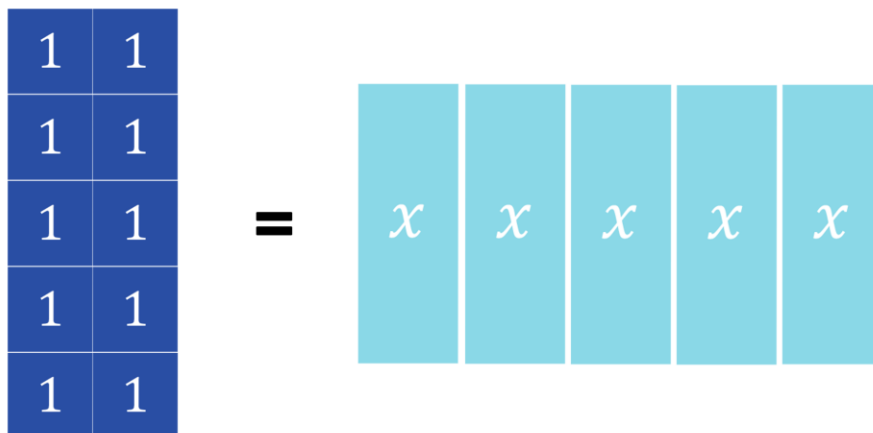
Step	Algebra Tiles	Description
Step 1		Use algebra tiles to represent the equation $-15 = 3x$.
Step 2		
Step 3		

2. Complete the table using algebra tiles to sketch the solution of $\frac{y}{4} = 5$ and explain each step. The first row has been completed.

Step	Algebra Tiles	Description
Step 1		Use algebra tiles to represent the equation $\frac{y}{4} = 5$.
Step 2		
Step 3		

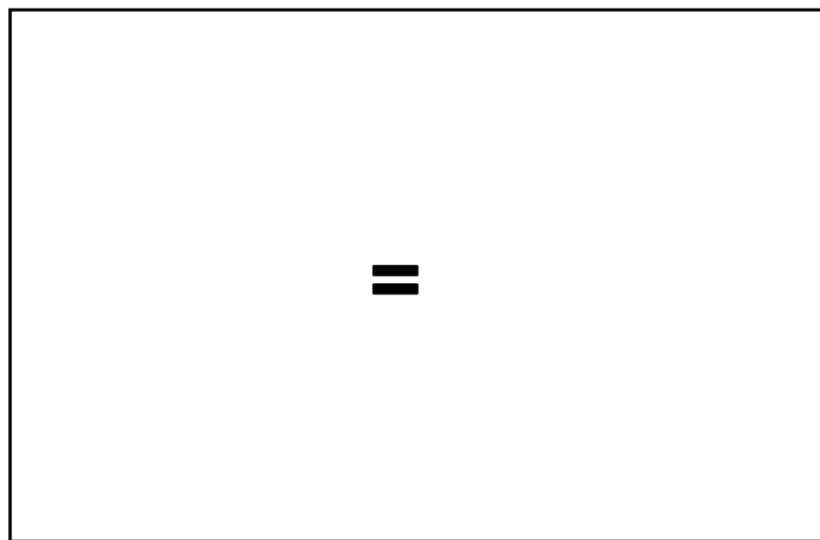
12.10.5 Your Turn!

1. The equation $10 = 5x$ is modeled on the equation mat.



Miranda told her classmate the value of x is $\frac{1}{2}$ because $5 \div 10 = \frac{1}{2}$.

- a. Explain whether you agree with Miranda.
- b. Sketch the next step in determining the solution to the equation.

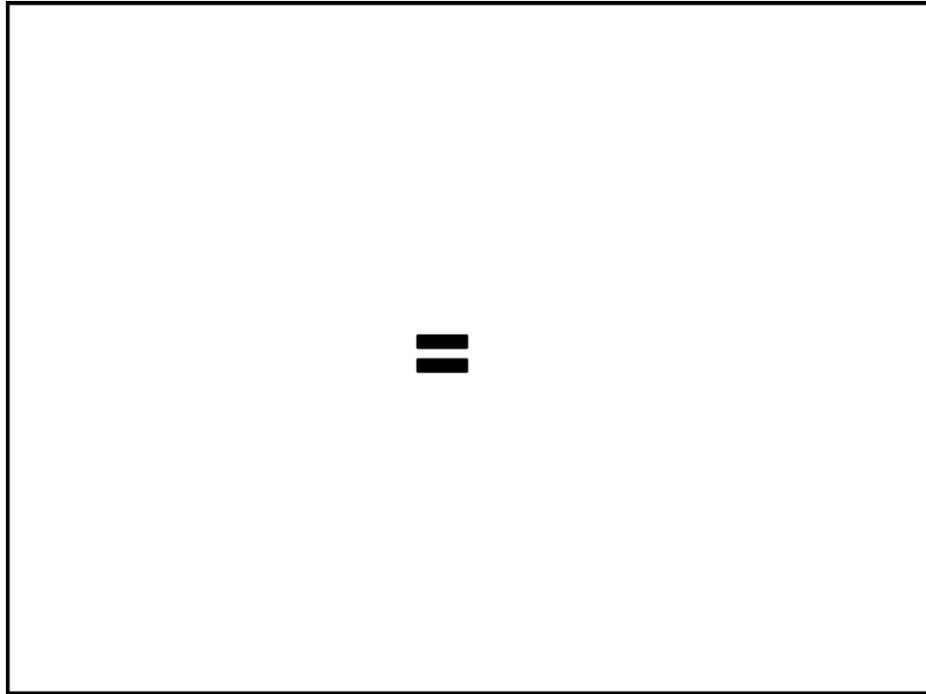


- c. What is the value of x according to your sketch?

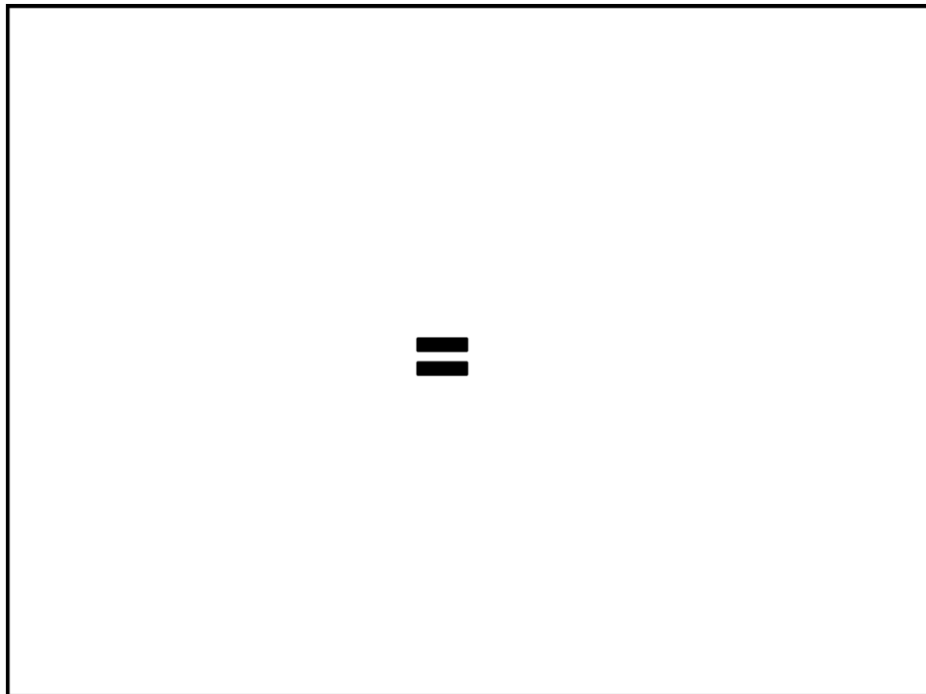
2. Consider the equation $\frac{x}{7} = -2$.

a. Using algebra tiles, sketch each step to find the solution of $\frac{x}{7} = -2$.

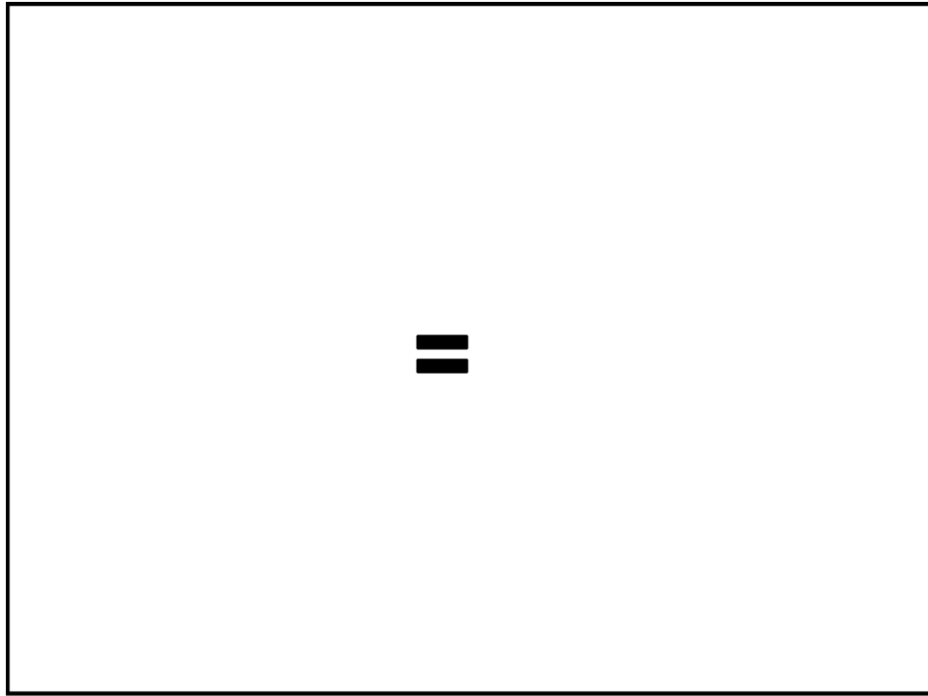
Step 1



Step 2



Step 3



- b. What is the value of x according to your sketch?

12.10.6 Wrap-Up: Cool Down

1. Create a sales pitch to explain inverse operations.
2. Share your sales pitch with a partner in 60 seconds or less.